

Yash Tyagi

Full-Stack Engineer · Real-Time Systems · IoT

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EDUCATION

KCC Institute of Technology and Management

B.Tech — Computer Science & Engineering

Greater Noida, UP

2023 – 2027

Coursework: DSA, OS, DBMS, Computer Networks, Software Engineering

PROJECTS

VolunteerCompass | [Live](#) | [GitHub](#)

2026

React 18, TypeScript, Node.js, Express, PostgreSQL, Prisma, Socket.io, Redis, Bull, JWT, Zod, Vercel, Render

- Architected full-stack volunteer platform with Leaflet maps, RSVP workflows, and RBAC (Volunteer / Organizer) serving 200+ active users across 15+ live events
- Cut match-suggestion latency 60% via Bull/Redis async queue; Socket.io notifications deliver updates under 80ms; CI/CD on Vercel + Render with zero-downtime Prisma migrations across 6 entity types

Sync Home — IoT Home Automation | [Live](#) | [GitHub](#)

2026

React, TypeScript, Vite, Three.js (R3F), GSAP, MQTT, TanStack Query, Tailwind CSS, Recharts, Vitest, Playwright

- Led frontend in a 4-person team; built real-time MQTT dashboard controlling 10+ smart devices with sub-200ms response time, architected for seamless hardware scaling
- Delivered 3D home visualization via React Three Fiber + GSAP with live IoT telemetry — 98% Lighthouse score, 94% unit test coverage (Vitest + Playwright), ~70% reduction in post-deploy bugs

AI Governance & Cost Management Platform | [GitHub](#)

2026

Next.js 14, TypeScript, Go, Python, PostgreSQL, ClickHouse, Kafka, Redis, Docker, Kubernetes, OPA (Rego), FastAPI, Terraform, AWS EKS

- Designed multi-tenant SaaS proxy (11 microservices) intercepting AI API calls across Claude, GPT-4, Gemini & Azure OpenAI — real-time cost tracking and policy enforcement under 50ms added latency
- Kafka + ClickHouse pipeline handles 50K+ events/min; OPA (Rego) enforces DLP, rate limiting & approval workflows; ARIMA + XGBoost ensemble forecasts 30-day spend at 92% accuracy with Isolation Forest anomaly detection

ACHIEVEMENTS

- Top 5 — Hackaccino 4.0: Top 5 out of 2,936 teams at Bennett University's 24-hour national hackathon — ₹15L+ prize pool — Feb 2026
- Top 5 — Mind Installers Hackathon 4.0: Top 5 at IIMT College of Engineering's 30-hour national hackathon, Greater Noida — Apr 2026
- HackIndia Spark 4: Competed nationally among 10,000+ registered participants across India
- Solved 150+ LeetCode problems, building a strong foundation in data structures and algorithms
- Cricket Team Captain: Led 11-member team to championship victory at KCC Institute's institutional cricket tournament, 2023-24

CERTIFICATIONS

JPMorgan Chase & Co. — Software Engineering Job Simulation · Forage · Jan 2026

Completed practical tasks in: Project Setup, Kafka Integration, H2 Integration, REST API Integration & REST API Controller

TECHNICAL SKILLS

Languages Java, JavaScript (ES6+), TypeScript, Go, Python, PHP, C++, HTML5, CSS3, SQL

Frontend React 18, Next.js, Three.js, GSAP, Tailwind CSS, shadcn/ui, Redux, Vite

Backend Spring Boot, Node.js, Express.js, FastAPI, Laravel, Socket.io, REST APIs, MQTT, Kafka, JWT, Zod

Database PostgreSQL, ClickHouse, MongoDB, Redis, Prisma ORM

DevOps Docker, Kubernetes, Terraform, AWS EKS, Git, GitHub CI/CD, Vercel, Render

Testing Vitest, Playwright

Concepts DSA, OOP, MVC, Microservices, Real-Time Systems, IoT